

HIGH-YIELD REVISION SHEET: MOLECULAR BASIS OF INHERITANCE

Rapid Revision Snapshot • Formulae, Key Metrics, Landmark Proofs & Operon Flowcharts

1. Structural Dimensions & Quantitative Rules (DNA)

Genome Metrics & Size Standards

- **φ×174 Bacteriophage:** 5,386 nucleotides (ssDNA)
- **Bacteriophage Lambda (λ):** 48,502 bp (dsDNA)
- **E. coli Genome:** 4.6×10^6 bp (~1.36 mm length)
- **Human Genome (Haploid, n):** 3.3×10^9 bp
- **Human Genome (Diploid, 2n):** 6.6×10^9 bp (~2.2 meters)

Watson-Crick B-DNA Parameters

- **Helical Diameter:** 2.0 nm (20 Å) — uniform
- **Pitch per turn:** 3.4 nm (34 Å) with 10 bp/turn
- **Distance between consecutive bp:** 0.34 nm (3.4 Å)
- **Bond Types:** N-Glycosidic (Base-Sugar), Phosphoester (Sugar-PO₄), 3'-5' Phosphodiester, H-bonds (A=T: 2; G≡C: 3)

Chargaff's Rule: $[A] + [G] = [T] + [C]$ | $[A]/[T] = 1$ | $[G]/[C] = 1$ | $([A]+[T])/([G]+[C]) = \text{Constant}$ per species

2. DNA Packaging: Prokaryotes vs Eukaryotes

Nucleosome Structure (Eukaryotes)

- **Histone Core:** Octamer (2 each of H2A, H2B, H3, H4) rich in basic amino acids **Lysine & Arginine**.
- **Core DNA:** ~200 bp wrapped 1.65 turns around octamer.
- **Linker Histone (H1):** Seals entry/exit of DNA strand.
- **Higher Levels:** 11 nm (Beads-on-string) → 30 nm (Solenoid) → Chromatin loops anchored by **NHC proteins**.

Euchromatin vs Heterochromatin

Feature	Euchromatin	Heterochromatin
Staining / Density	Light, loosely packed	Dark, tightly packed
Activity	Active (Transcribes)	Inactive (Silent)
Replication	Early S-phase	Late S-phase

3. Landmark Experiments: Unequivocal Proofs

Experiment	Scientist(s) & Year	Experimental System	High-Yield Conclusion
Transforming Principle	Frederick Griffith (1928)	<i>Streptococcus pneumoniae</i> (virulent Smooth 'S' vs avirulent Rough 'R') in mice.	Heat-killed S transformed live R cells into virulent S; established a genetic transforming agent.
Biochemical Nature	Avery, MacLeod, McCarty (1933–44)	Purified S-strain lysates treated with Protease, RNase, and DNase.	Transformation abolished <i>only</i> by DNase; proved DNA is the chemical transforming factor.
Unequivocal Proof	Hershey & Chase (1952)	Bacteriophage T2 with ³⁵ S (protein coat) vs ³² P (DNA core) infecting <i>E. coli</i> .	Only ³² P entered bacterial pellet; ³⁵ S stayed in supernatant. Proved DNA is genetic material.
Semi-Conservative Replication	Meselson & Stahl (1958)	<i>E. coli</i> grown in ¹⁵ NH ₄ Cl transferred to ¹⁴ N; analyzed via CsCl centrifugation.	Gen 1 (20 min): 100% Hybrid (¹⁵ N- ¹⁴ N); Gen 2 (40 min): 50% Light, 50% Hybrid. Taylor et al. confirmed in <i>Vicia faba</i> using ³ H-thymidine.

4. DNA Replication Fork & Transcription Units

Replication Machinery

- **DNA Polymerase:** Catalyzes only in 5' → 3' direction (~2,000 bp/sec rate in *E. coli*).
- **dNTPs:** Dual role — act as substrates & hydrolyze pyrophosphate for energetic drive.
- **Leading Strand:** Continuous along 3' → 5' template.
- **Lagging Strand:** Discontinuous Okazaki fragments joined covalently by **DNA Ligase**.

Transcription Unit Essentials

- **Template Strand:** 3' → 5' polarity (read by RNA Pol).
- **Coding Strand:** 5' → 3' polarity (matches mRNA except U replaces T; reference for upstream/downstream).
- **Bacterial Factors:** **σ (Sigma)** for initiation; **ρ (Rho)** for termination.

Eukaryotic RNA Polymerases & Processing (Post-Transcriptional Modifications)

- **RNA Pol I:** Transcribes 28S, 18S, and 5.8S rRNAs.
- **RNA Pol II:** Transcribes precursor mRNA / hnRNA.
- **RNA Pol III:** Transcribes tRNA, 5S rRNA, and snRNA.
- **Splicing:** Spliceosome excises non-coding *introns* and ligates coding *exons*.
- **5' Capping:** Addition of 7-methylguanosine triphosphate (m⁷Gppp).
- **3' Tailing:** Polyadenylation (200–300 adenylate residues).

5. Genetic Code, tRNA Adapter & Translation

Genetic Code Cardinal Rules

- **Triplet & Commaless:** 64 codons read continuously.
- **Unambiguous:** 1 codon = only 1 amino acid.
- **Degenerate:** 1 amino acid can have multiple codons.
- **Dual AUG:** Start codon + codes for Methionine.
- **Stop Codons:** UAA (ochre), UAG (amber), UGA (opal). *No tRNA exists for stop codons.*

Translation & Peptidyl Transfer

- **Aminoacylation:** Amino acid + ATP → Aminoacyl-AMP → Aminoacyl-tRNA.
- **tRNA Structure:** Cloverleaf 2D (L-shape 3D); Anticodon loop + CCA-3' amino acid acceptor arm.
- **Ribozyme:** **23S rRNA** in bacteria (28S in eukaryotes) catalyzes peptide bond formation.

6. Lac Operon & Regulation (Jacob & Monod)

Operon Structure: [p (Promoter)] – [i (Regulator)] – [p (Promoter)] – [o (Operator)] – [z] – [y] – [a]

Gene	Encoded Product	Biological / Enzymatic Function
i gene	Lac Repressor Protein	Constitutively produced; binds operator (o) to block RNA Pol transcription.
z gene	β-Galactosidase	Hydrolyzes lactose into glucose and galactose; converts some to allolactose.
y gene	β-Galactoside Permease	Increases cell membrane permeability to entry of lactose.
a gene	β-Galactoside Transacetylase	Transfers acetyl group from acetyl-CoA to β-galactosides.

Absence of Inducer (Operon OFF)

Repressor protein binds tightly to operator (o) → RNA Polymerase blocked → Structural genes (z, y, a) NOT transcribed.

Presence of Inducer (Operon ON)

Lactose / Allolactose binds and inactivates repressor → Operator cleared → RNA Polymerase transcribes polycistronic lac mRNA.

7. Human Genome Project (HGP) & DNA Fingerprinting Flashcard Summary

HGP Quick Data Points

- **Genome Size:** 3.1647×10^9 bp (~30,000 total genes).
- **Protein Coding:** < 2% of the genome codes for proteins.
- **Gene Counts:** Chromosome 1 has most (2,968); Y has fewest (231).
- **Average & Largest Gene:** Avg = 3,000 bp; Largest = Dystrophin (2.4 Mb).
- **Polymorphism:** 1.4 million SNPs (Single Nucleotide Polymorphisms).

DNA Fingerprinting (Alec Jeffreys)

1. Isolate DNA & digest with Restriction Endonucleases.
2. Fractionate fragments by size via Agarose Gel Electrophoresis.
3. Southern Blot: Transfer single-stranded DNA to nylon membrane.
4. Hybridize with radiolabeled **VNTR probe** (Variable Number Tandem Repeats, 0.1–20 kb mini-satellites).
5. Detect individual-specific banding patterns via **Autoradiography**.